

Class 1 – Ch 4- Computer Programming

- Basic Programming Idea
- Variables
- Data types
- Function

MCCQ

✓ ✓ Example – 2577

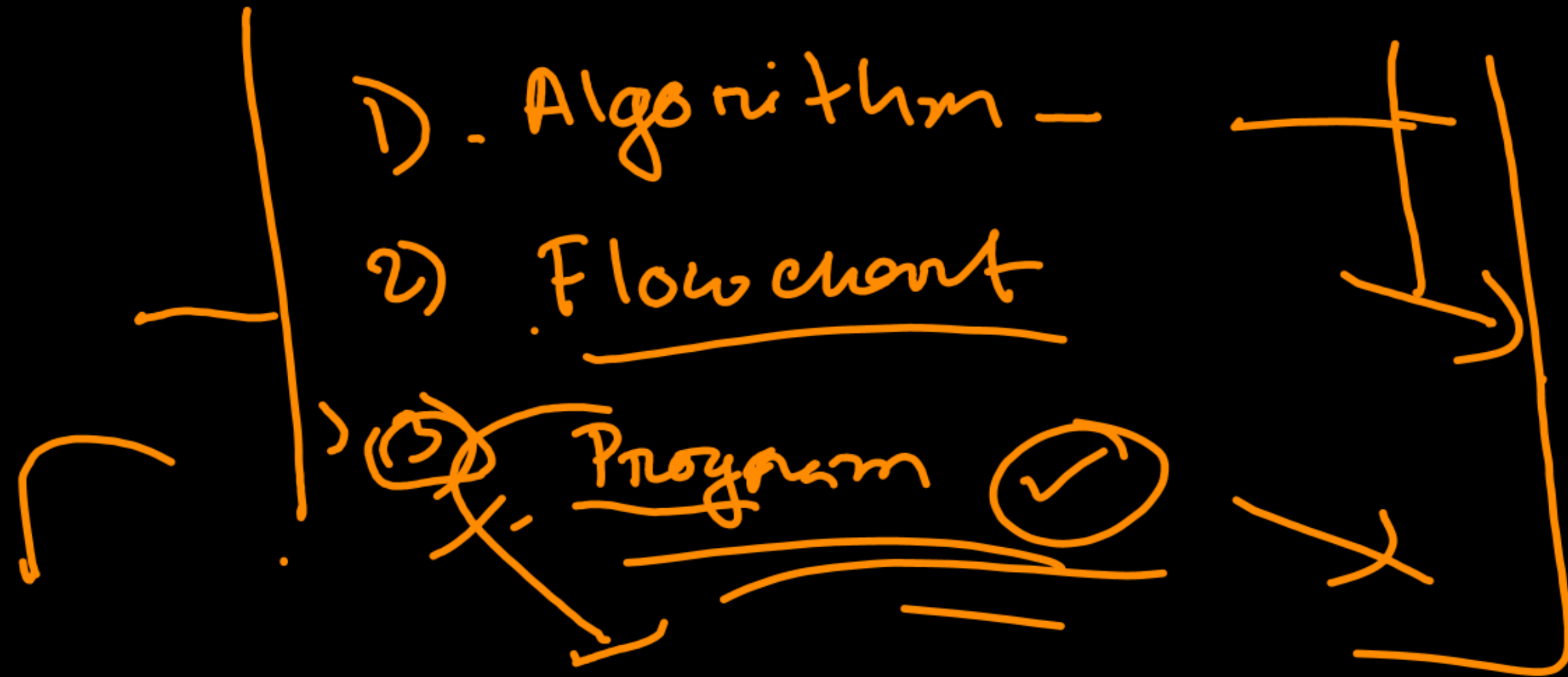
✓ 12345 → OOP

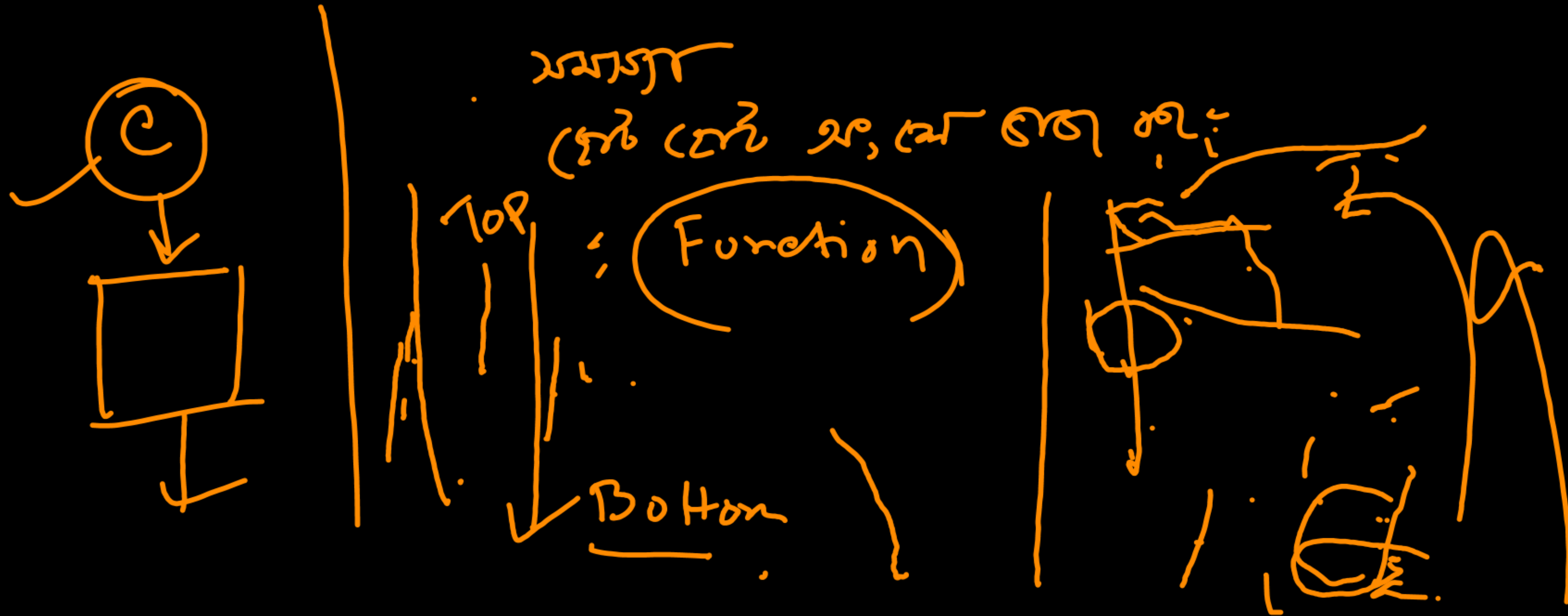
100
PDF

Problems solved,

Program :







```
int add ( )  
{  
    int a;  
    int b;  
}
```

```
int a;  
int main ( )  
{  
    |  
    |
```

Top - Down

- (Function)

→ code reuse

→ code reusability

→ Security

Bottom - Up

Object/class

- code reuse

→ code — Encapsulation

→ Encapsulation, Abstraction

Variable (ଆବଶ୍ୟକୀୟ) —

1) A-Z, 0-9, — ଆବଶ୍ୟକୀୟ ଅଟେ

2) First_Name ✓

3) କ୍ଷେତ୍ରର ଆବଶ୍ୟକୀୟ ଅଟେ

int name ✓ ✓

name = first X

2name ✓
{✓

name/. X

name2 ✓

✓ LOVE.
✓ love.

Arithmetic operator — $+$, $-$, $*$, $\div$

Assignment operator — $=$

Logical operator — $\&$, $\|$, $!$

Bitwise — $\&$, $\|$, $\wedge$, $\sim$

Unary

$++$, $--$

$\&$

$\>$

$\ll$

$\gg$

int a = 10; =
a += 5;

$$\begin{aligned}\Rightarrow a &= a + 5 \\ &= 10 + 5 \\ &= 15\end{aligned}$$

$$\begin{aligned}a &*= 5 \\ a &= a * 5 \\ a &= 10 * 5 \\ &= 50\end{aligned}$$

```
int a = 10, b = 20;  
printf("%d", a < b);
```

10 < 20

<

$a \leq b$

True → 1 ✓

False → 0

```
int a = 10, b = 30, c = 20;
```

```
printf("Logical AND: %d\n", (a > b) || (a > c));
```

Output → 0

10 > 30 ↓
F OR 10 > 20 - F

```
printf("Logical OR : %d\n", (c < b) || (b > c));
```

1.9

0

30 < 20 20 > 30
T F
1 - 1

int a = 5, b = 3; OR

printf("AND: %d\n", a & b);

printf("XOR: %d\n", a ^ b);

1011
1101

0110

0 1
1 0

1 1

= 6
4 + 2
1 1 0

T T = T

5 - 101
3 - 011

001
①

(ANS) 1 20

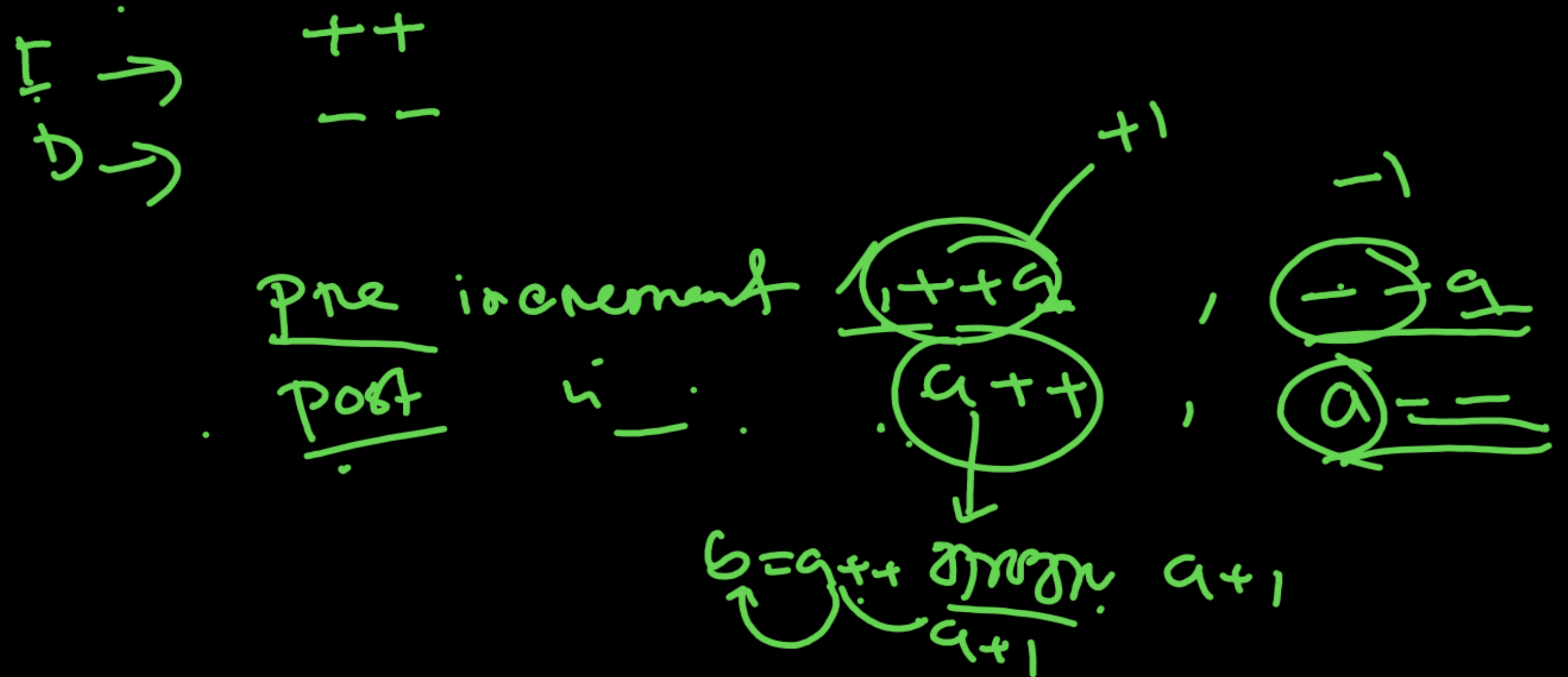
XOR = 110

OR =

$$\begin{array}{r} 1011 \\ 1101 \\ \hline 1111 \end{array}$$

Not

$$\& \sim 9 = \begin{array}{r} 1011 \\ \downarrow \downarrow \downarrow \downarrow \\ 0100 \end{array}$$



```
int a = 10;  
int b = a++;
```

$$\begin{array}{l} \underline{b} = \underline{10} \\ \underline{a} = 10 + 1 \end{array}$$

```
printf("a = %d\n", a);
```

```
printf("b = %d\n", b); → 10
```

$a = 10$
 $b = a++$
 $\rightarrow b = 10, a = 10 + 1 = 11$

$\checkmark$ $b_2 + 9$
 $\quad \quad \quad 181313$
 ② $\checkmark$ $b = 9 + 1 = 10$

— $a = 12$,
— $b = 20$,
— $c = 0$,
→ $a = b$ →
← $b = a$,
← $c = b$.

$a = 20$ →
 $b = 20$ →
 $c = 20$ →

$$a = 10$$

$$\textcircled{1+10}$$

$$= 11$$

$$\underline{1+a} = \underline{11}$$

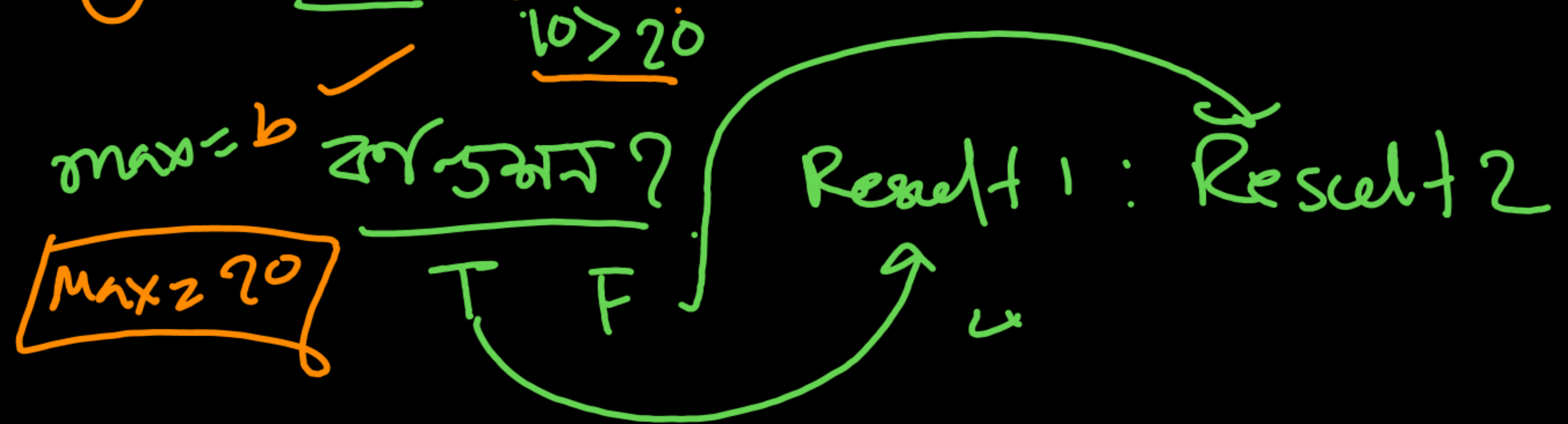
$$\textcircled{\underline{a+1}}$$

$$= \underline{10} + \underline{1} = 11$$

```
int a = 10, b = 20;
```

④

```
int max = (a > b) ? a : b;
```



- ✓ Basic — int, float, double, char, string
- ✓ User define Datatype — Union, Structure, —
- ✓ Derived Datatype — Array, Pointer,

int — 2 — 4 byte

%d
%f

int A[5] = { 20, 10, 5, 30, 100 }

$$A[0] = 20$$
$$\overline{A[0] + ? =}$$

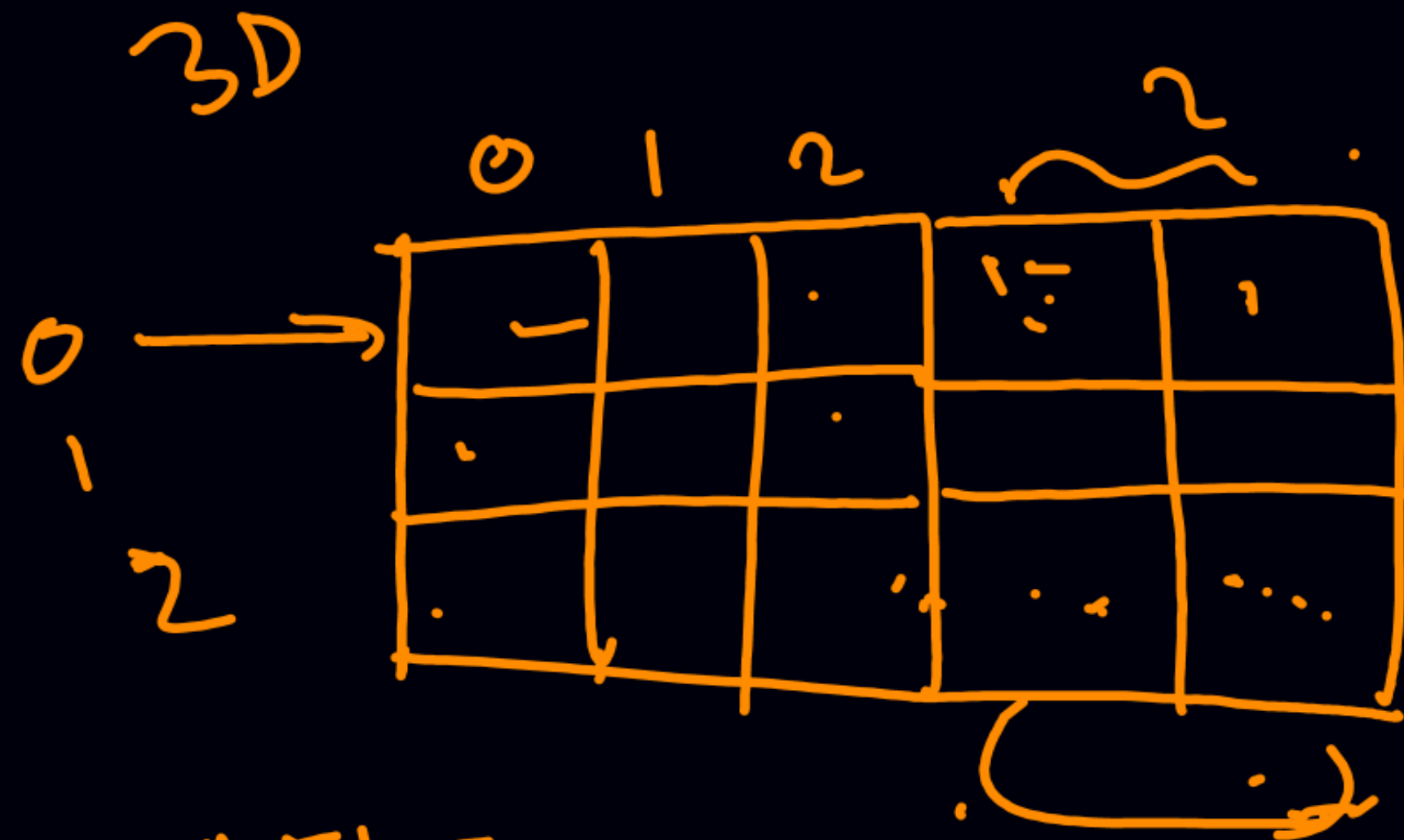
Two Dimensional Array

2 $A[2][3] = 0$

Row Col

0	1	2
9	2	3
1	0	5

$\text{print}(A[0][3])$



$A[3][3][2]$

2 ~~*~~

a = 10

*P = 2a

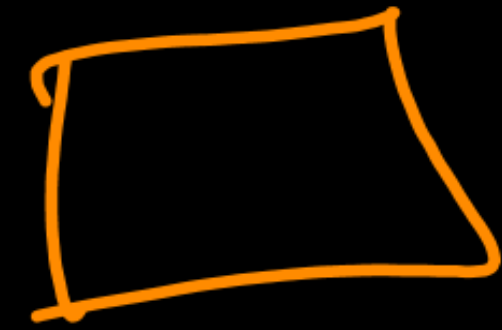
P = 10

a = 10 + 10
= 20

..

Structure

Union



Struct Student

```
int Roll;
float avg;
char[20];
```

Memory

```
struct student st;
(1, 3.50, "Suman")
printf "Student s
- - - Roll"
```

enum {
 Jan = 0 ✓
 Feb = 1
 Mar = 2
 | 3
 | 4
 | 5
 | 6
 | 7
 | 8
 | 9
 | 10
 | 11
 | 12
}

Remnant Month = Jan
..

- 1) user Define
- 2) Bilt in. function / library Function

print search , power , fon

```

int Sumam() {
    int a = 10;
    int k = 20;
    return a * b;
}

```

Print = sumam();

printf("%d", sumam());

Output 30

What will be the output of the following C code? [COMBINED BANK-SO-IT-24]

```
int main() {
```

```
    int i = 11, j = 3;
```

```
    printf("%d\n", i | j);
```

```
    return 0;
```

```
}
```

ক) 0 খ) 5 গ) 6 ঘ) 11

Bitwise
OR

1011
0011

1011 → 11

What will be the output of the following C code? [COMBINED BANK-SO-IT-24]

```
int x = 20, y = 10, z = 5;  
printf("%d", x > y > z);  
return 0;
```

Handwritten note: 20 > 10 > 5

✓ क) 1 ख) 0 ग) 10 घ) 20

What will be the output of the following C code? [COMBINED BANK-SO-IT-24]

```
int main() {  
    float p = 10.5; ✓  
    int a = 5 * p + 5.0; ✓  
    printf("%d\n", a);  
    return 0;
```

✓

57

$$\begin{aligned} & 5 * 10.5 \\ &= 52.5 + 5 \\ &= \underline{57.5} \end{aligned}$$

What will be the output of the following C code?
[AP, COB-23]

```
struct xx {  
    int x=3;  
    char name[] = "hello";  
};
```

Handwritten notes: A green bracket on the left side of the struct definition. A green circle around 'xx' with an arrow pointing to 'int x=3;'. A green circle around 'x=3' with an arrow pointing to 'char name[] = "hello";'. A green circle around 'name[]' with the word 'Size' written below it. A green circle around 'xx' with an 'X' inside. A green box around '()' with an arrow pointing to 'struct'.

- A. 3 hello B. Compiler Error
C. Linking error D. None of these

What will be the output of the following C code?

[AP, COB-24]

✓ int y = 0; 1
int x = (y != 0);
printf("%d", x);

Y = 0

T = 1
F = 0

Y = 5

✓ A. 0 ~~B. 1~~

C. Negative number D. Compiler error

row, col
row
int data[2][3][2] = { { {1, 2}, {3, 4}, {5, 6}},
{ {7, 8}, {9, 10}, {11, 12} }

What will be the output of data[0][2][1]? -

[SO IT CoB-24]

A. 0 B. 5 C. 6 D. 11



- *What is programming? Give 4 examples [ADA, SB - 2026]*
- *Define problem solving technique. [45 BCS, Computer Science]*
- *Write the differences between structured programming and object-oriented programming. [45th BCS CSE 2024; SO, CoB - 2021; AP ICT ministry-2021, ADA, SB - 2026]*
- *Write down the advantages of OOP over traditional structured programming language. [Officer IT, CoB - 2023, JID-10152]*